



MPhil in Data Intensive Science

Submission deadline: 24th June 2026

GW minor module - Coursework Assignment

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*The coursework will be submitted via a GitLab repository which will be created for you. You should write a report of no more than 3000 words to accompany the software you write to solve the problem. The approximate number of words that should be dedicated to explaining each part of the problem is indicated in square brackets; e.g. [x]. You should place all your code and your report in this repository. The report should be in PDF format in a “**report**” folder. You will be provided access to the repository until the deadline above. After this point, you will lose access to the repository, which will constitute submission of your work.*

The software associated with the coursework should be written in Python and follow best software development practice as defined by the Research Computing module. It should be possible for someone else to easily run your code to reproduce all of the results included in your report.

Quasinormal modes and black hole ringdown

Black holes (BHs) are remarkably simple objects. As Chandrasekhar put it, “The black holes of nature are the most perfect macroscopic objects there are in the universe: the only elements in their construction are our concepts of space and time.” The *no-hair theorem* implies that all stationary (astrophysical) BHs are fully characterised by just two quantities: the mass, $M > 0$, and angular momentum, $J = \chi M^2$ with $0 \leq \chi < 1$.

Kerr BHs are stable. When perturbed they oscillate as they settle back down; a process known as *ringdown*. The complex frequencies of these oscillations, $\omega_{\ell mn} = 2\pi f_{\ell mn} - i/\tau_{\ell mn}$, known as *quasinormal modes* (QNMs), form a discrete spectrum labelled by indices $\ell \geq 2$, $-\ell \leq m \leq \ell$, and $n \geq 0$. (Here we consider only with *prograde* QNMs.) The QNM frequencies are functions of the spin and scale inversely with the mass,

$$\omega_{\ell mn}(M, \chi) = \frac{\hat{\omega}_{\ell mn}(\chi)}{M}.$$

This formula is in natural units with $G = c = 1$. The dimensionless functions $\hat{\omega}_{\ell mn}(\chi)$ are provided in the `qnm` package¹. The QNM model for the ringdown signal is given by

$$h(t) = \sum_{\alpha} A_{\alpha} \cos(2\pi f_{\alpha}(t - t_0) + \phi_{\alpha}) \exp\left(\frac{-(t - t_0)}{\tau_{\alpha}}\right),$$

where α is a compound index identifying a particular mode (e.g. $\alpha = (\ell, m, n)$), A_{α} is the mode amplitude and ϕ_{α} is the mode phase. This model applies only after the ringdown start time, $t > t_0$.

GW astronomy allows us to observe the ringdown. The final stages of a BH merger can be modelled as a perturbation of the remnant BH. Identifying QNM frequencies in the ringdown allows us to infer the BH parameters, test the consistency of general relativity and the Kerr metric, and measure the area of the remnant BH’s event horizon², all with a minimum of detailed theoretical modelling.

Analysing the ringdown presents several challenges in addition to those usually encountered in GW data analysis. Because we only want to analyse the ringdown, it is first necessary to identify a start time t_0 and then to cut the data at this time so only times $t > t_0$ are included in the analysis. Performing a sharp cut like this where the signal is expected to be large leads to problems if the analysis is performed in the frequency domain. This motivates us to find ways of analysing GW data in the time domain.

¹The `qnm` package can be found on GitHub.

²The area of the event horizon is an important theoretical quantity because it is proportional to the entropy of the BH.

Data files

The following files are published with this coursework:

- `data.dat`. This contains 32s of data from a single interferometer sampled at 4096Hz. The data has been bandpass filtered using a 4th order Butterworth filter between 20 and 400Hz. A loud BH merger occurs near the middle of the data. The first column contains the time (s) and the second contains the strain (dimensionless).
- `psd.dat`. This contains the power-spectral density (PSD) for the instrumental noise in the data. The first column contains the frequency (Hz) and the second contains the PSD $S_n(f)$ (Hz^{-1}).

Coursework Project

1. Consider the following two (prograde) QNMs: $(\ell, m, n) = (2, 2, 0)$ and $(3, 3, 0)$. Calculate the dimensionless frequencies using the `qnm` package. Plot the real and imaginary parts of these frequencies as a function of χ . [100]

Load the strain data from the file `data.dat`. You should split the data into off-source data (which doesn't contain any GW signal and which will be used to estimate the noise properties) and on-source data (which contains the GW data) stating your choices clearly. You should further identify a time t_0 in the analysis data which is a suitable ringdown start time; only on-source data with $t > t_0$ will be analysed in this project. You should think carefully about the choice you make for t_0 and justify your choice clearly in your report (you might also consider experimenting with different choices).

2. Plot all 32s of the strain data showing clearly your chosen values for the start and end of the on-source data and the ringdown start time.

Throughout this coursework you should use the QNM ringdown model for $h(t)$ given above where the sum is restricted to include only two modes. Your model should allow the frequencies and damping times of the two modes to be independent parameters. This model has 8 free parameters: $\theta = (A_1, \phi_1, f_1, \tau_1, A_2, \phi_2, f_2, \tau_2)$.

First, consider the white-noise (WN) log-likelihood,

$$\log \mathcal{L}_{\text{WN}}(d|\theta) = -\frac{1}{2} \sum_i \frac{(d(t_i) - h(t_i; \theta))^2}{\sigma^2} + \text{constant}.$$

3. Using the off-source data, estimate a suitable value for σ . [100]

(TURN OVER)

4. Sample the posterior distribution for the model parameters θ . You should state clearly what priors you use. Plot the posteriors on the QNM frequency and damping time parameters. (You might find that your posteriors contain multiple, disconnected secondary peaks. It will simplify the following analysis if you focus on the largest peak in the posterior.) [100]

Questions 5 to 7 get you to consider to what extent this WN noise likelihood is appropriate for ringdown analysis.

5. Using the off-source data, estimate the frequency-domain power-spectral density (PSD) using a Welch periodogram. Plot the result. [100]
6. From the PSD, estimate the time-domain noise autocorrelation function $C(t - t')$. Plot the result. [100]
7. Explain how the value you estimated for σ in part 3 relate to your estimate for $C(t - t')$ in part 6. [100]

For the final part of this coursework, question 8, you should use the posterior samples you obtained using the WN likelihood in question 4.

8. Describe how these results can be used to perform a consistency test of the theory of general relativity and the Kerr metric. Perform a consistency test on this signal and comment on the result. [100]
9. Discuss (in under 200 words) why observations of merging black holes with gravitational waves are a good place to test the theory of general relativity? [100]